

Homework 7

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the $\text{\LaTeX}$ solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due date which can be found on the course website: <https://ecealgo.com/su26/homeworks.html>.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the “assigned” date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due is a recipe for failure.

Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you *must* cite *every* source that you use, and you *must* write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

See the course web site (<https://ecealgo.com/su26>) for more information.

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

Extra Instructions Solutions to a dynamic programming problem have (at minimum) three things:

- A recurrence relation
- A *brief* description of what your recurrence function represents and what each case represents.
- A *brief* description of the memory element/storage and how it's filled in.

1. You are given a directed graph $G = (V, E)$ with positive length edges, as well as two vertices s and t . An edge $e \in E$, is considered bad if the cost of all walks from s to t that uses e costs at least 3β , where β is the length of the shortest path from s to t . Describe an algorithm that computes all the *bad* edges in G . Slower algorithms would earn 60% of the total points.
2. You are given the following mystery recurrence. Your task is to determine what it computes. Once understood, provide a one-sentence English description of the recurrence SP and analyze its, memoized, most optimized runtime.

Recurrence Definition:

Base Case 1:

$$SP(u, v, k) = 0 + M(u, u, k) \quad \text{if } u = v \text{ and } k \geq 0$$

Base Case 2:

$$SP(u, v, k) = \infty + M(u, u, k) \quad \text{if } k < 0$$

Recursive Case:

$$SP(u, v, k) = \min \left(SP(u, v, k-1) + M(u, u, k), \min_{(x,v) \in E} [SP(u, x, k-1) + w(x, v) + M(u, u, k)] \right) + M(u, v, k)$$

$$M(u, v, k) = \begin{cases} M(u^2 + 1, v^3 + 6, k-1) & \text{if } k > 0 \\ 0 & \text{if } k \leq 0 \end{cases}$$